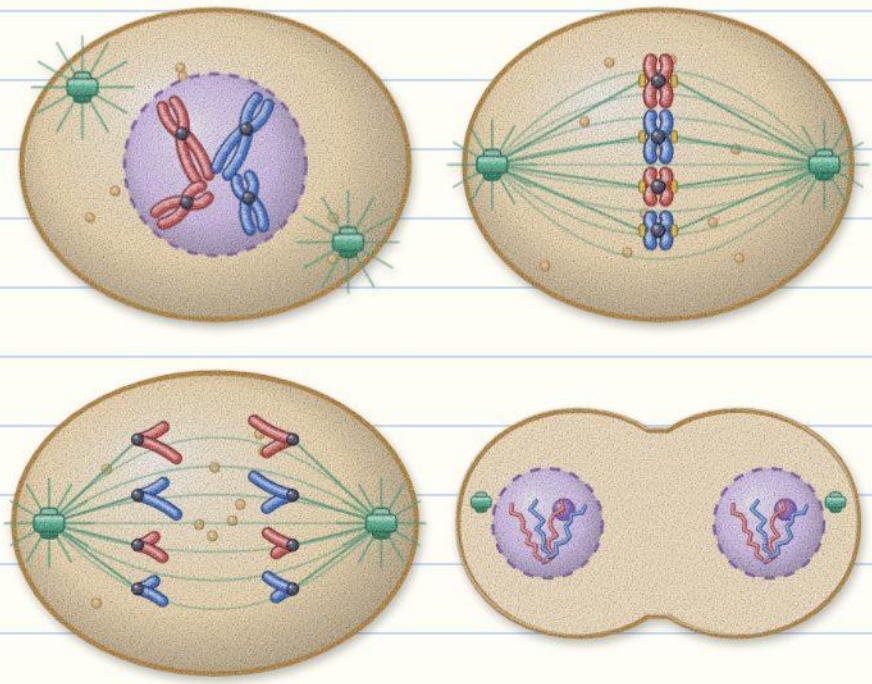
Cell Cycle & Cell Division

PU-I Biology · Chapter 10

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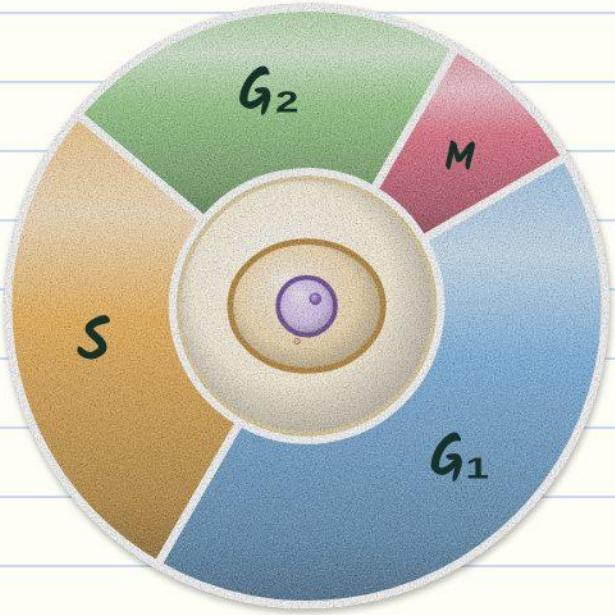
★ = very important for exams

I. The Cell Cycle

- ★ Cell cycle = sequence of events: cell duplicates its genome, makes other cell parts and divides into 2 daughter cells.
- Events are under genetic control. Cell growth is continuous but DNA synthesis only in S phase.
- ★ Duration: human cell ≈ 24 h, yeast ≈ 90 min.

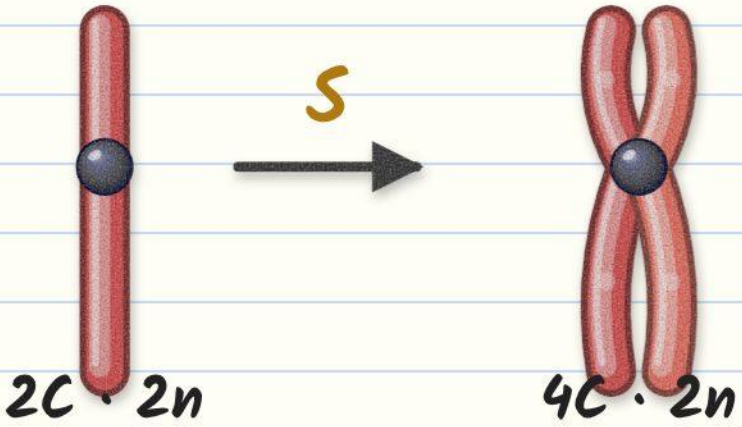
Two phases

- Interphase – preparation, > 95% of the cycle ("resting" but very active!)
- M phase – about 1 h: karyokinesis (P → M → A → T) + cytokinesis



2. Interphase

- G_1 – between mitosis & DNA replication; cell grows, no DNA replication.
- ★ S – DNA replication: 2C → 4C but chromosome no. stays 2n. Centriole duplicates (animal cells).
- G_2 – proteins for mitosis are made; growth continues.
- ★ G_0 (quiescent) – cells exit G_1, active but not dividing (e.g. heart cells).



Stage	Chromosomes	DNA
G_1	2n	2C
G_2	2n	4C
after mitosis	2n	2C
after meiosis I	n	2C
after meiosis II	n	C

Animals: mitosis only in diploid somatic cells · plants: haploid & diploid cells.

3. Prophase & Metaphase

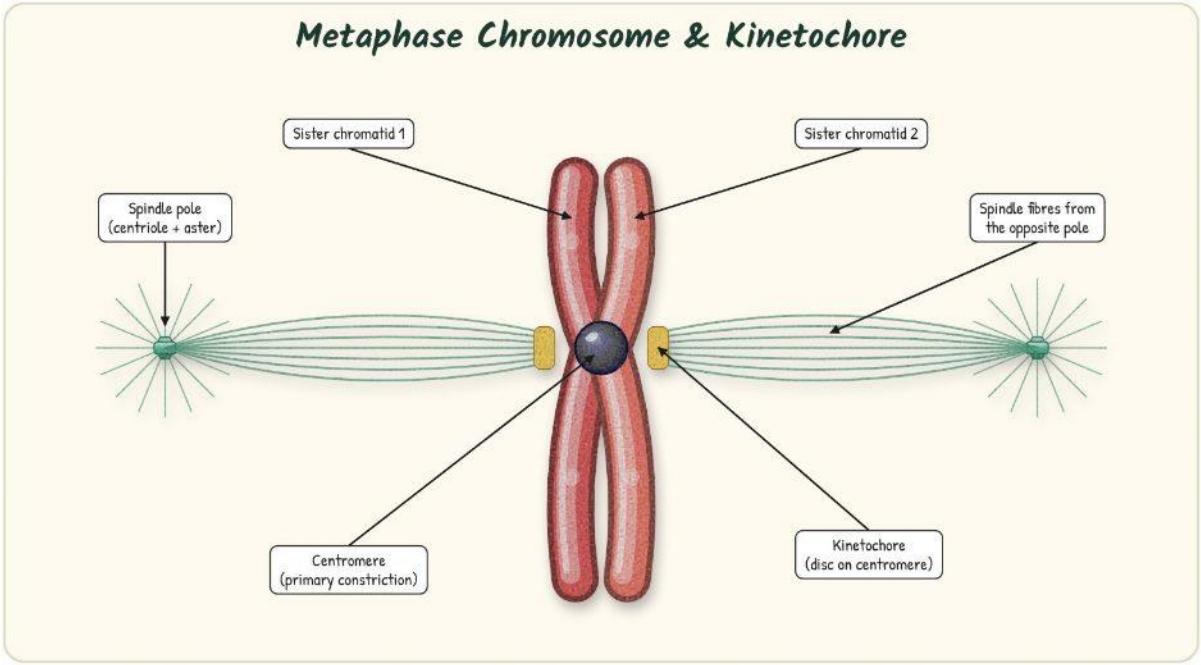
★ Mitosis = equational division (same chromosome no.). Strasburger (1870, plants), Flemming (1879, animals, coined "mitosis").

Prophase

- Chromatin condenses → chromosomes with 2 chromatids joined at centromere.
- Centrosomes move to opposite poles; asters + spindle = mitotic apparatus.
- Golgi, ER, nucleolus, nuclear envelope disappear.

Metaphase

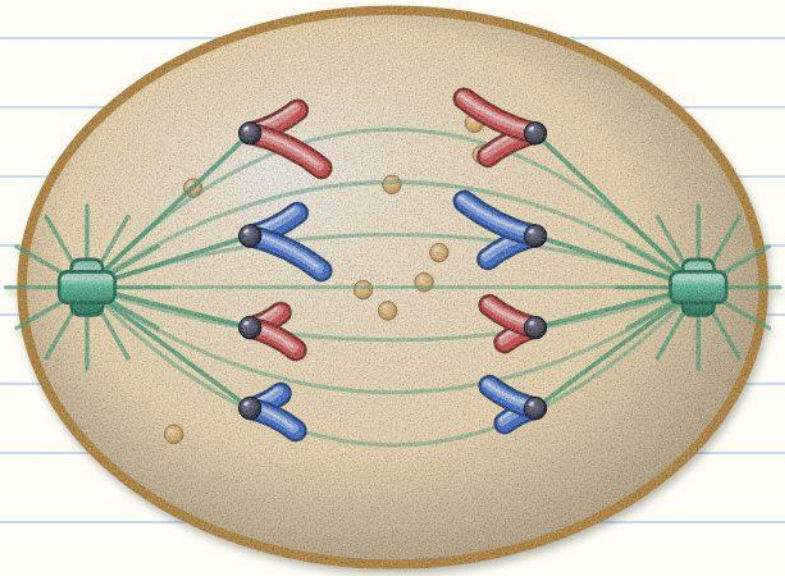
- Nuclear envelope fully gone; condensation complete.
- ★ Kinetochores (discs on centromeres) attach spindle fibres.
- ★ Chromosomes line up on the metaphase plate — best stage to study chromosomes.



4. Anaphase & Telophase

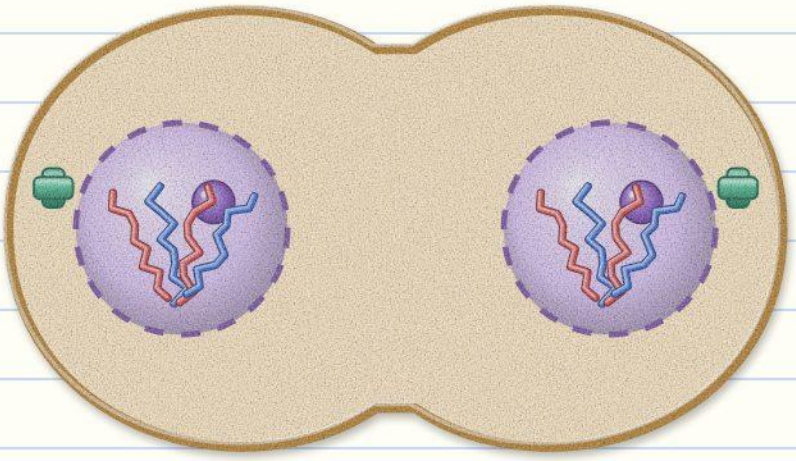
Anaphase

- ★ Centromeres split; chromatids move to opposite poles.
- Centromere leads, arms trail behind (V / J shapes).



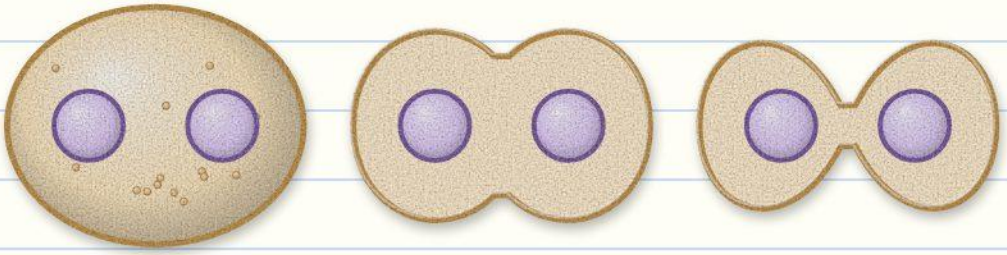
Telophase

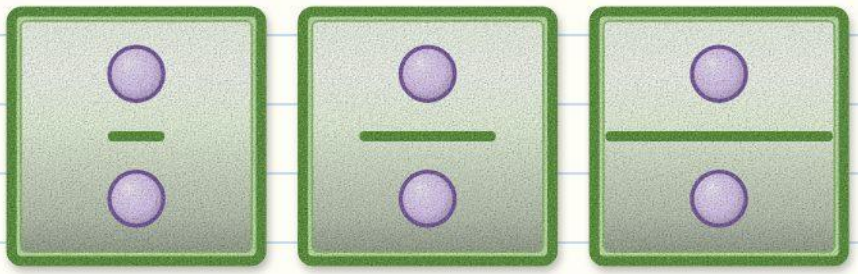
- Chromosomes reach poles, decondense, lose individuality.
- Nuclear envelope forms → 2 daughter nuclei; nucleolus, Golgi, ER reform.
- ★ Telophase = reverse of prophase.



5. Cytokinesis

- Animal cell: furrow deepens & joins in the centre (outside → in).
- ★ Plant cell: cell plate forms in the centre and grows outwards; cell plate = middle lamella.
- ★ No cytokinesis after karyokinesis → syncytium (coconut liquid endosperm).





Significance of mitosis

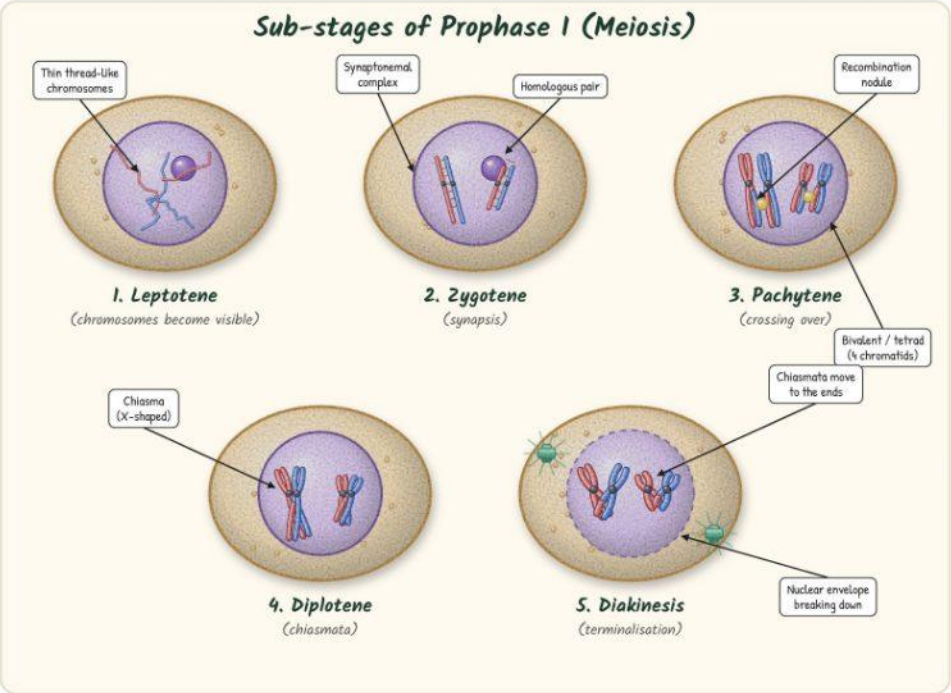
- Diploid, identical daughter cells · growth · repair (epidermis, gut lining, blood cells)
- Restores nucleo-cytoplasmic ratio · apical & lateral meristems · vegetative reproduction

6. Meiosis & Prophase I

- ★ Reductional division: 2 divisions, 1 DNA replication → 4 haploid cells.
- Pairing of homologues + crossing over between non-sister chromatids.

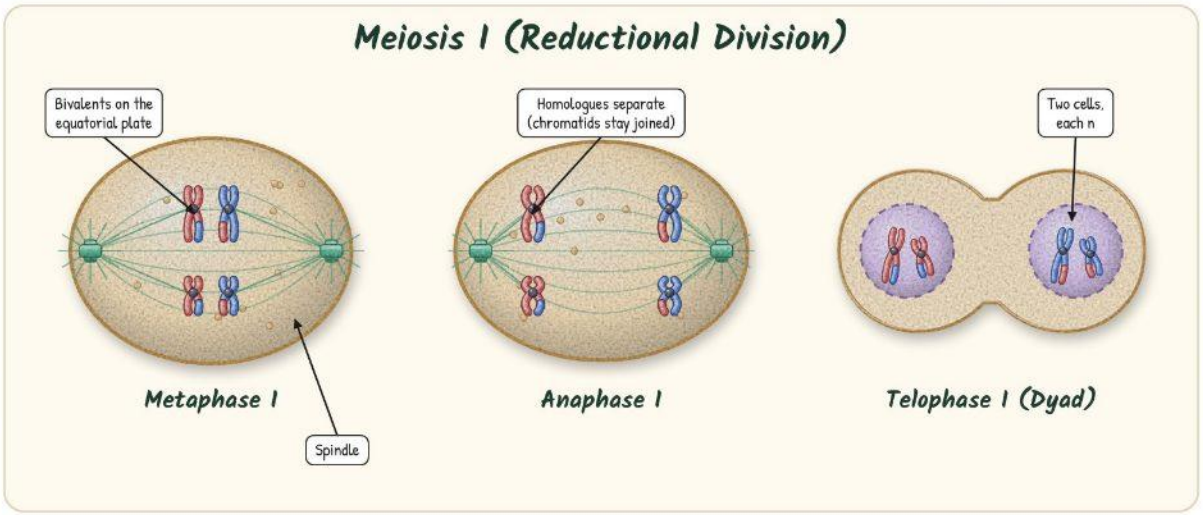
"Lazy Zebras Play During Day"

- Leptotene – chromosomes visible
- ★ Zygotene – synapsis, synaptonemal complex, bivalent
- ★ Pachytene – crossing over by recombinase
- ★ Diplotene – complex dissolves, chiasmata (months/years in oocytes)
- Diakinesis – terminalisation; nucleolus & envelope vanish



7. Meiosis I

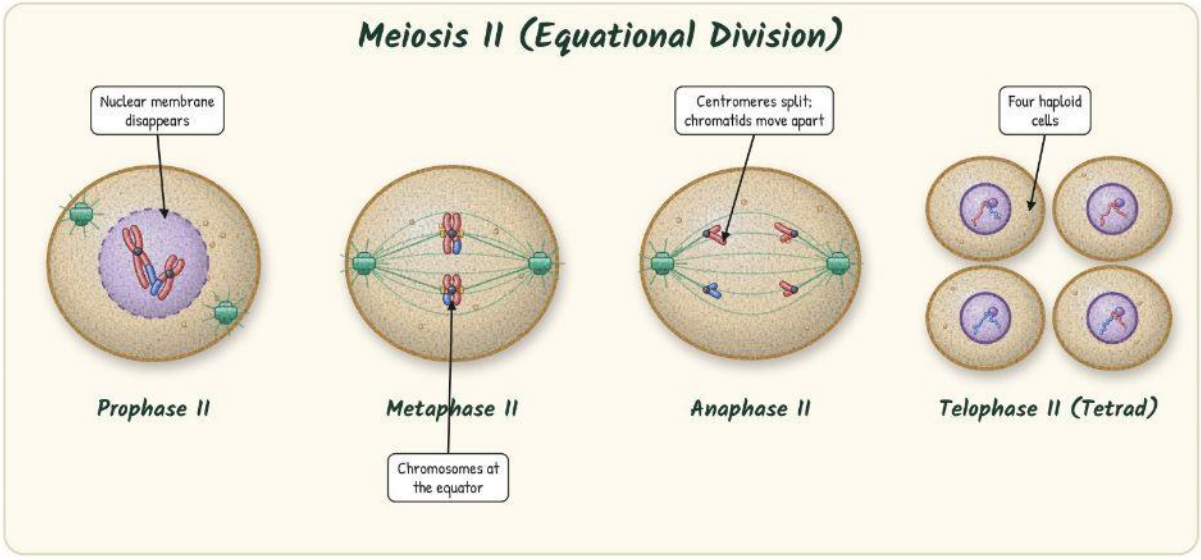
- Metaphase I — bivalents on the equatorial plate.
- ★ Anaphase I — homologues separate, sister chromatids stay joined → 2n becomes n.
- Telophase I — envelope & nucleolus reappear → dyad.
- ★ Interkinesis — short, no DNA replication.



	Mitotic anaphase	Anaphase I
Separates	sister chromatids	homologues
Centromere	splits	does not split
Type	equational	reductional

8. Meiosis II & Significance

- Meiosis II resembles mitosis (equational).
- Metaphase II — fibres to kinetochores of sister chromatids.
- ★ Anaphase II — centromeres split.
- Telophase II — tetrad = 4 haploid cells.



Significance of meiosis

- ★ Keeps chromosome number constant across generations.
- Crossing over → variation → raw material for evolution.

★ Exam Corner

Feature	Mitosis	Meiosis
Divisions	1	2
Daughter cells	2, diploid	4, haploid
Crossing over	no	yes
Type	equational	reductional + equational
Where	somatic cells	gametogenesis

NEET numbers

- 100 pollen → 25 meioses · 100 seeds → 125 meioses
- 2n = 20 → 10 bivalents → 40 chromatids in pachytene
- 46 chromosomes at metaphase → 92 chromatids

